Design and Implementation of an AI-Based Hand Gesture Recognition System using IMU Sensor and KNN Algorithm

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***Abstract*—This paper gives the design and implementation of an artificial intelligence (AI)-based hand gesture recognition system that can facilitate intuitive human-computer interaction via the use of inertial sensing technology. The wearable glove provides a method to record and capture the data of the hand movements by using an inertial measurement unit (IMU), in this case MPU6050. The IMU measures multi-axis acceleration and angular velocity that is associated with various hand gestures. The obtained motion data is in the form of multidimensional fea- ture vectors and is calculated with the help of a microcontroller based system. K-Nearest Neighbours (KNN) machine learning algorithm is used to categorize the gesture data that has been captured into pre-determined categories of gestures. When the gestures are identified, they are coded to the relevant control commands and sent to a computing device through either wired communication interface or wireless communication interface. The proposed method also avoids reliance on cameras and lighting conditions, unlike the vision-based gesture recognition system, and thus the approach will be more robust, portable, and flexible in its operation. The system provides a low-cost, scalable and small-sized solution that can be used in the field of human-computer interaction, assistive technologies, and smart device control as well as interactive systems.**

***Keywords*—Artificial Intelligence (AI), Hand Gesture Recognition, Inertial Measurement Unit (IMU), MPU6050, K-Nearest Neighbors (KNN), Machine Learning, Human–Computer Interaction (HCI), Wearable Glove Interface, Motion Classification, Embedded Systems.**

I. Introduction

Human-Computer Interaction (HCI) has developed much during the last decades as more natural and intuitive means of interaction are replacing traditional input devices, like keyboards and mice. Among these innovations, the hand gesture recognition has turned out to be one of the strong methods that can allow users to interact with computing systems by means of physical gestures. Gesture interfaces offer more engaging and intuitive form of interaction especially in applications where other traditional input devices are not practical or efficient.

Traditional gesture recognition systems are mostly opting of vision based methodologies involving cameras and image processing methods. Despite the fact that such systems have proven to be very accurate, they are also limited in a number of ways such as being sensitive to lighting conditions, background complexity, limitations in positioning the camera and excessive computer power. These shortcomings lower their usability in the field and limit their mobility. In order to defeat these barriers, gesture recognition systems through sensor-based recognition have been receiving growing attention because of their strength and their independence on the environmental factors.

The alternative to vision-based systems is an inertial sensing technology, especially when using Inertial Measurement Units (IMUs). An IMU is a combination of acceleration and gyroscopes, which determine the linear acceleration and angular velocity in various axes. The MPU6050 is used in this work as the fundamental sensing element. The sensor is incorporated in a wearable glove that will record the dynamic movements of the hand. The system can record the multiple axis acceleration and angular velocity data to generate motion signatures of various predefined gestures.

Embedded system that uses microcontroller processes the captured sensor data and transforms into the raw readings into structured multidimensional feature vectors. A machine learning classifier takes as inputs these feature vectors. K-Nearest Neighbors (KNN) algorithm is chosen among the other methods of classification because of its simplicity, effi- ciency, and the use of small-to-mediocre data sets. The KNN algorithm categorizes the received gesture information in terms of comparing the information with already stored training data and identifying the matching information which is closest in terms of distance measures like the Euclidean distance.

The identified gestures are then converted to pre-defined control commands and sent to an external computing device either via a wired or wireless communication interface. This allows real time control of applications, smart devices and interactive systems. Some of the benefits of the proposed system will be low cost, portability, scalability, and external environmental free nature.

The main task of the given work is to develop and introduce an effective AI-based hand gesture recognition system on the basis of inertial sensing and machine learning methods. The purpose of the system is to offer an effective solution to applications in the field of human-computer interaction, assistive technology, rehabilitation system, gaming interface, and smart device control. The proposed solution removes the need to rely on the inputs based on the vision, thus, increasing the flexibility of its operation and increasing its applicability to various real-life settings.

II. Literature Review

The idea of hand gesture recognition has been widely researched within an area of Human-Computer Interaction (HCI) as it may offer a natural and intuitive interaction of human and computer. Current methods of gesture recognition can be categorized into sensor based methods and vision based methods. Gesture recognition systems that are based on vision use cameras and image processing algorithms to analyze and identify gestures of hands. Gesture classification has been conducted by many researchers using depth cameras, RGB cameras in combination with machine learning and deep learning models like Convolutional Neural Networks (CNNs). An example of this is the Microsoft Kinect used as a hand tracking and gesture recognition device. Even though these systems exhibit high classification accuracy, they are characterized by a number of limitations such as light sensitivity, background noise, occlusion, and high computation. Moreover, the reliance on external cameras make portability and limits application in the outdoor or dynamic setting.

Gesture recognition systems based on sensors have received enormous interest in order to overcome these challenges. These systems use wearable technology that is of flex sen- sors, accelerators, gyroscopes, and Inertial Measurement Units (IMUs) to track motion data directly inspired by hand movements. Sensor-based systems offer a better robustness, reduce computational complexity and they are not dependent on the environmental lighting conditions as compared to vision-based techniques. IMU-based systems have been shown to be very useful in the capture of dynamic hand gestures among sensor-based systems. The MPU6050, a unit combining both a 3 axis gyroscopic sensor and an accelerator, has been popular in gesture recognition studies because of its low price, small size and the consistency with which it can track 3 axis movement. IMU data has been used to obtain motion features of acceleration magnitude, angular velocity, and change of orientation, which the researchers used to classify the data.

Gesture classification algorithms that have been used in different machine learning algorithms include Support Vector Machines (SVM), Decision Trees, Artificial Neural Networks (ANN), and K-Nearest Neighbors (KNN). Although deep learning methodologies are highly accurate, they demand large datasets, a lot of computing power, and a lot of training time. Conversely, the K-Nearest Neighbors algorithm is straightforward but at the same time, provides a reasonable classification model that can be used in small to medium sized datasets. It works on the distance measures like Euclidean distance and classifies by the majority vote among the K-Nearest neighbors. KNN due to its simplicity, ease of implementation and good performance in real-time embedded application is thought to be suitable in microcontroller based gesture recognition sys- tem. It has been established in recent studies that it is possible to combine IMU sensors with machine learning algorithms to create portable and wearable gesture recognition systems. Yet, there are several current implementations that are based on either the sophisticated computational models or restricted sets of gestures. There are also those that do not have scalability or real-time responsiveness.

The limitations discussed in this paper are tackled in the proposed system that combines a wearable IMU-based sensing apparatus with a small scale KNN classification algorithm that is run on an embedded platform. The proposed solution does not have any limitations to the environment like changes in light and background interference like in camera-based systems. Moreover, the system is portable, cost-effective, and scalable, which makes it possible to use it in the assistive technologies, control of smart devices, gaming interfaces, and interactive systems.

III. Methodology

The approach of the research hand gesture recognition system based on AI involves a procedure of organised steps such as data collection, data preprocessing, features selection, gesture recognition with the K-Nearest Neighbors (KNN) algorithm and command mapping. The general goal of such methodology is to properly detect pre-programmed hand gestures within the framework of the inertial sensing technology whilst maintaining the performance in real time and portability of the given system.

A. System Overview

The proposed system combines a wearable glove which has an Inertial Measurement Unit (IMU), a processing unit with a microcontroller, and a machine learning classifier. The sensor used to measure the movement is the IMU sensor, which is the MPU6050 attached on the glove, to record the data as a result of hand movements. The sensor is three-axis linear acceleration (Ax, Ay, Az) and three-axis angular velocity (Gx, Gy, Gz), which give the six dimensional motion information. The obtained sensor data are send to a microcontroller where preliminary processing is made and the organized data are sent to a computing device to be classified.

B. Data Acquisition

There are a variety of hand gestures that the user is able to make during the data acquisition stage when he is wearing the glove. The IMU sensor measures the values of the acceleration and angular velocity direction along the X, Y and Z axis, in relation to every gesture. These are read out at constant frequency to keep up the temporal resolutions constant. The instances of gestures are captured at a fixed time interval, and several samples are taken to form a training set. The data obtained are stored as time-series vectors of the motion patterns of each gesture class.

C. Data Preprocessing

Raw IMU sensor data can be more noisy and may be have small variations as a result of tremors in the hands or other environmental distractions. Preprocessing is thus carried out to enhance quality of data. This stage includes:

- Reduction of noise by means of smoothing.
- Scaling sensors data to uphold consistent scale.
- Elimination of outliers, when it is required.

The raw data is preprocessed and then organized into the multidimensional feature vectors and ready to be fed up into a machine learning classifier.

D. Feature Extraction

The extraction of features is very important in enhancing the accuracy of classification. The time-series data is already pre-processed then the appropriate statistical and motion features are extracted. These may include:

- Average acceleration in each axis.
- Standard deviation.
- Maximum and minimum values.
- Signals magnitude vector.
- Changes in angular velocity.

These parameters are what are extracted into a small feature vector that describes every sample. The classification algorithm takes the feature as input.

E. KNN Classification of Gestures

K-Nearest Neighbors (KNN) algorithm is used to classify gestures because it is simple and effective when dealing with small and medium-size data. KNN is built on a distance measurement principle of feature space. The training phase involves the storage of labeled feature vectors of predefined gestures in memory. During the testing stage, as a new gesture is done:

- The system derives its feature.
- Euclidean distance between the new vector and all the training vectors that are stored is computed.
- The K neighboring closest are selected.
- These neighbors are divided into majority on which the gesture is categorized.

KNN K value is experimentally selected to tradeoff accuracy in classification with computational efficiency.

F. Command Mapping and Communications

After the identification of a gesturing, a gesture is mapped to a specified control command. These commands can be related to actions like moving a cursor, controlling a device or application-specific actions. The command signals created are sent to a computing device, via wired (USB serial communication) or wireless communication interfaces. This mapping process allows the interaction of real-time between the user and the outside system.

G. System Implementation Flow

The general approach process could be summarized into the following sequential steps:

- Movement of hands carried out by user.
- The IMU sensor records the motion information.
- Microcontroller gathers and preprocesses data.
- The feature vectors are created.
- The gesture is classified using KNN algorithm.
- Gesture that is known is mapped to control command.
- The command is relayed to computing device.

IV. Hardware Implementation

The hardware design of the proposed hand gesture recognition system is such that it will be portable, reliable and real-time. The device has a wearable sensing unit, a processing unit, a wireless communication unit and a power supply unit.

A. Inertial Measurement Unit (IMU)

The MPU6050 will be attached to a glove that the wearer wears to record the movement of the hands. It incorporates a 3-axis accelerator and a 3-axis gyroscope, giving the ability to provide the six-dimensional motion data (Ax, Ay, Az, Gx, Gy, Gz). The sensor interacts with the Arduino Uno by the use of the I2C protocol via the SDA and SCL pins.

B. Microcontroller Unit

The system contains the Arduino Uno as the central processing unit. It retrieves sensor data, does basic preprocessing, and prepares the data to be examined further. The reason why the Arduino Uno is selected is because the board is easy to program, has enough memory to run embedded applications, and can be integrated with different sensor and communication modules.

C. Bluetooth Communications Module

The HC-05 Bluetooth Module provides the hardware and computing device to be in wireless communication with one another. It works on UART Serial communication via the TX and RX pins. Gesture data and the control command that is recognized are sent in real time by the Bluetooth module.

D. Power Supply Unit

The battery is a 12V rechargeable battery that is attached to the Arduino Uno power input. The Arduino includes onboard voltage regulator that supply 5V of power needed to power the IMU sensor and the Bluetooth module.

E. Hardware Integration and Working

The hardware connection is organized in the following way:

- MPU6050 connected to Arduino through I2C interface (SDA, SCL).
- HC-05 attached to Arduino through UART interface (TX, RX).
- Arduino power input connected to the battery 12V.

The IMU captures motion information. Arduino processes and formats sensor readings and sends them through Bluetooth to the computing device to be classified with the help of the KNN algorithm.

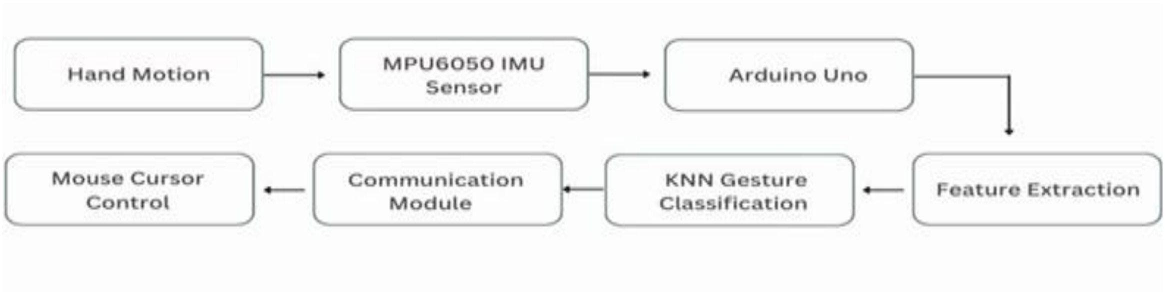

Fig 1:- Block Diagram of AI-Based Hand Gesture Recognition System using IMU Sensor and KNN Algorithm

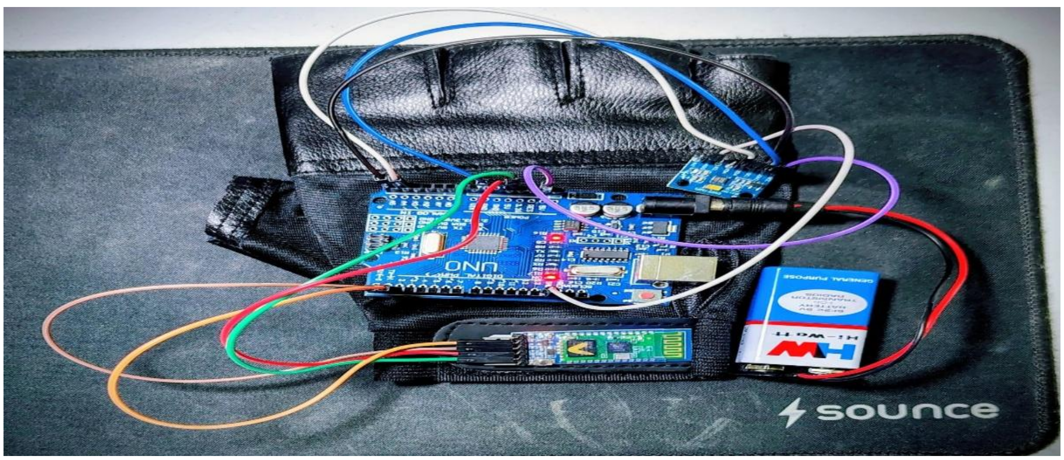

Fig 2:- Hard ware Implementation of AI-Based Hand Gesture Recognition System using IMU Sensor.

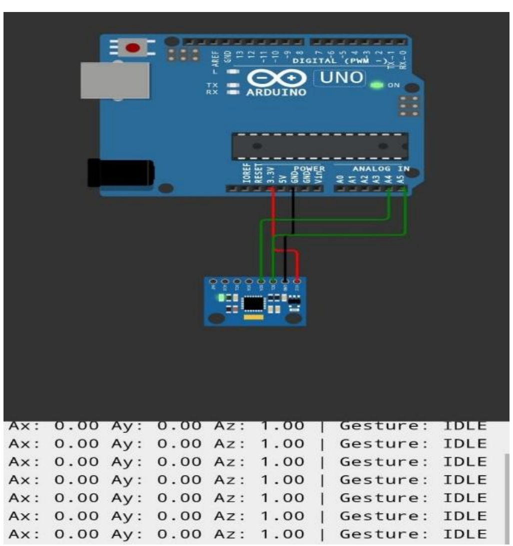

Fig 3:- Simulation Output of IMU sensor and Arduino (“Gesture : IDLE” because the sensor is in rest).

V. Simulation and Results

A. Simulation Setup

The hardware components were Arduino Uno, MPU6050 and HC-05 Bluetooth Module. The IMU sensor was attached to a glove. Sensor data were sent to the computer through Bluetooth for KNN classification.

B. Data Collection and Testing

Pre-programmed gestures included: Open hand, Closed fist, Left, Right, Up, and Down movements. These were repeated several times to obtain the dataset. The gathered data were split into training and testing samples.

C. Results

1) **Accuracy:** In normal conditions, the system was successful in classification of the predefined gestures.
2) **Response Time:** The system reacted fast, giving almost real-time results.
3) **Robustness:** The system did not require cameras, proving useful in varied lighting environments.

D. Discussion

The findings indicate that the IMU based method is a good gesture recognizer that is cheap and easy to implement. The system is simple to train and appropriate for small datasets. However, with more complicated gestures, accuracy can slightly reduce. Future developments can involve more efficient feature extraction or enhanced machine learning algorithms.

VI. Conclusion

A hand gesture recognition system based on inertial sensing technology using AI has been designed and implemented. The system uses the MPU6050 attached on a wearable glove to record motion data in the form of multi-axis acceleration and angular velocities. The data acquired are analyzed with the Arduino Uno and classified via the KNN algorithm. Experimental findings confirm the system is good in terms of accuracy and almost real-time performance. In contrast to camera-based systems, it enhances resilience and portability.

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